BOAMP Successor Linkage Methodology
Current Benchmark State

BOAMP Data Science Internship Project

2026-08-13T21:25:33

Technical Summary

This report states the current defensible BOAMP successor-linkage pipeline. The study does not claim to certify legal contract renewals. Its event is an *observable successor procurement*: a later BOAMP procurement episode from the same buyer that is sufficiently similar to an earlier awarded digital procurement episode.

The current Grand Ouest survival cohort contains 3,800 awarded digital procurement episodes. The main linkage rule accepts 544 successor links, giving an observed event rate of 0.143 and a median observed successor time of 31.82 months among linked events. The candidate pool is broad: 763,417 candidate pairs from 3,520 anchors with at least one candidate.

The current national benchmark contains 252 labelled anchors and 7,031 labelled anchor-candidate rows. On the validation split, `M_B_text_ranking @ 0.70` remains the strongest precision-first choice: precision 0.800, recall 0.182, false-positive rate 0.000, and 5 accepted validation links. `M_C_weighted_gated` recovers more positives (recall 0.409) but admits more false positives (FPR 0.105). This trade-off is important because a false positive in a survival dataset creates both a false event and a false event time, while an abstained link is handled conservatively as right-censoring.

1 What The Pipeline Measures

Let i index an earlier awarded procurement episode and j index a later candidate episode. The object of interest is not a legal renewal certificate, because BOAMP notices do not provide that ground truth consistently. Instead, the pipeline estimates whether the data show a later procurement episode that is credible enough to be treated as a successor:

$$Y_i = \begin{cases} 1, & \text{if one accepted observable successor is found before the cutoff,} \\ 0, & \text{otherwise.} \end{cases}$$

The event time is

$$\tau_i = C_{\hat{j}} - A_i,$$

where A_i is the award-date origin of episode i, $C_{\hat{j}}$ is the first-publication date of the accepted successor, and $\hat{j}$ is the selected candidate. If no accepted successor is found before 2025-12-31, the episode is right-censored.

2 End-to-End Pipeline

The implemented pipeline is:

BOAMP notices → standardised notices → procurement episodes
→ candidate pairs → linkage scoring → one successor or abstention → survival dataset.

Standardisation and feature engineering. The preprocessing stage standardises dates, buyer identifiers, buyer names, CPV codes, text fields, procedure information, framework flags, and duration fields where they are explicitly available. It does not impute missing duration or invent expected expiry dates. Buyer standardisation is deliberately conservative: municipal variants such as commune/ville/mairie can be harmonised, but intercommunal legal forms are preserved as distinct entities, and conflicting validated SIRENs block a buyer match. The current legal-form audit reports 0 accepted links with conflicting validated SIREN evidence and 0 accepted municipal/intercommunal mixes.

Episode reconstruction. BOAMP is notice-level data, but the analysis needs procurement-level events. Therefore notices belonging to the same procurement are reconstructed into an episode. This avoids treating a consultation notice, correction notice, award notice, or lot-level administrative notice as separate renewals.

Candidate generation. Candidate generation is intentionally broad and is not the final model. For an anchor episode with award date A_i and candidate publication date C_j, the candidate is exposed only if

$$A_i + 90 \leq C_j \leq A_i + 2920.$$

The lower bound removes very near follow-up notices and parallel administrative activity. The upper bound is approximately eight years; it keeps the candidate pool inclusive enough for long public-procurement cycles. Precision is then controlled by the selection stage rather than by a narrow time window. The current blocking rule generated 763,417 candidate pairs, with a median of 89 candidates per anchor.

3 Current National Benchmark Reference

The current benchmark is national rather than Grand Ouest-only. It samples awarded digital procurement episodes across French region groups and CPV divisions. The dev split has 134 anchors and 3,638 pair rows; the validation split has 60 anchors and 1,763 pair rows. Only the probability frame supports national weighted estimates; enrichment frames are used for stress cases and diagnostics.

The current annotation caveat remains important: labels are protocol-generated development evidence with double-pass validation and adjudication. They are not official legal renewal ground truth and should not be described as human inter-annotator ground truth.

4 Linkage Algorithms

All methods operate on the same exposed candidate set. This is crucial: the primary method is not comparing text over the whole BOAMP universe. Buyer and time plausibility are imposed before text ranking.

M_A**: deterministic evidence.** Define B_{ij} as buyer-identity support, P_{ij} as CPV continuity, and T_{ij} as text similarity. The deterministic rule accepts a link only when strong buyer evidence is present, CPV continuity is positive, and a minimum text signal is present:

$$B_{ij} = 1, \quad P_{ij} > 0, \quad T_{ij} \geq t_A.$$

It is interpretable, but it loses recall when CPV or buyer identifiers are missing or noisy.

M_B**: text ranking.** For each episode, the text fields are converted to TF–IDF vectors x_i and x_j. Text similarity is cosine similarity:

$$T_{ij} = \cos(x_i, x_j) = \frac{x_i \cdot x_j}{\|x_i\|\|x_j\|}.$$

Within the same-buyer, plausible-time candidate set, the method selects

$$\hat{j}_i = \arg\max_j T_{ij},$$

and accepts the selected candidate only if

$$T_{i\hat{j}_i} \geq 0.70.$$

This is the primary method because it is simple, reproducible, auditable, and best matches the precision-first objective.

M_C**: weighted gated score.** This method combines evidence components into a score:

$$S_{ij} = 0.50B_{ij} + 0.25T_{ij} + 0.20P_{ij} + 0.05G_{ij},$$

where G_{ij} is a timing-plausibility score. Missing components are handled by the implemented score-normalisation logic rather than imputation. The method then applies gates and ranks by S_{ij}. It is useful as a higher-recall contrast, but the validation results show a materially higher false-positive rate.

M_D**: Fellegi–Sunter probabilistic linkage.** Let γ_{ij} be the discretised comparison vector for buyer, text, CPV, and timing evidence. Fellegi–Sunter estimates the likelihood of observing γ_{ij} under a match class M and a non-match class U, then uses the log-likelihood ratio

$$w_{ij} = \log \frac{P(\gamma_{ij} \mid M)}{P(\gamma_{ij} \mid U)}.$$

The fitted model provides `fs_match_weight` and `fs_match_probability`. In the current data it does not outperform the simple text-ranking rule, likely because true successors are rare and same-buyer procurement activity contains many non-renewal lookalikes.

M_E**: expiry-aware audit arm.** The expiry-aware method runs in parallel with the primary method. It estimates an expected end date E_i only from explicit end-date or duration evidence. It does not assume a four-year duration. Candidate timing is

$$R_{ij} = C_j - E_i.$$

Very early candidates require stronger text evidence and CPV continuity. This arm accepts 504 links, compared with 544 under the main method. It is retained as an audit/sensitivity check, not promoted, because observed market behaviour shows many declared successors can be published before expected expiry.

5 Dev Results

Method	Threshold	Accepted	Precision	Recall	FPR	Coverage
M_A_deterministic	70.000	10	0.600	0.128	0.035	0.075
M_B_text_ranking	70.000	24	0.875	0.447	0.035	0.179
M_C_weighted_gated	70.000	47	0.596	0.596	0.126	0.351
M_D_fellegi_sunter	65.000	4	0.000	0.000	0.023	0.030

Table 1: Unweighted dev metrics on all labelled current benchmark frames.

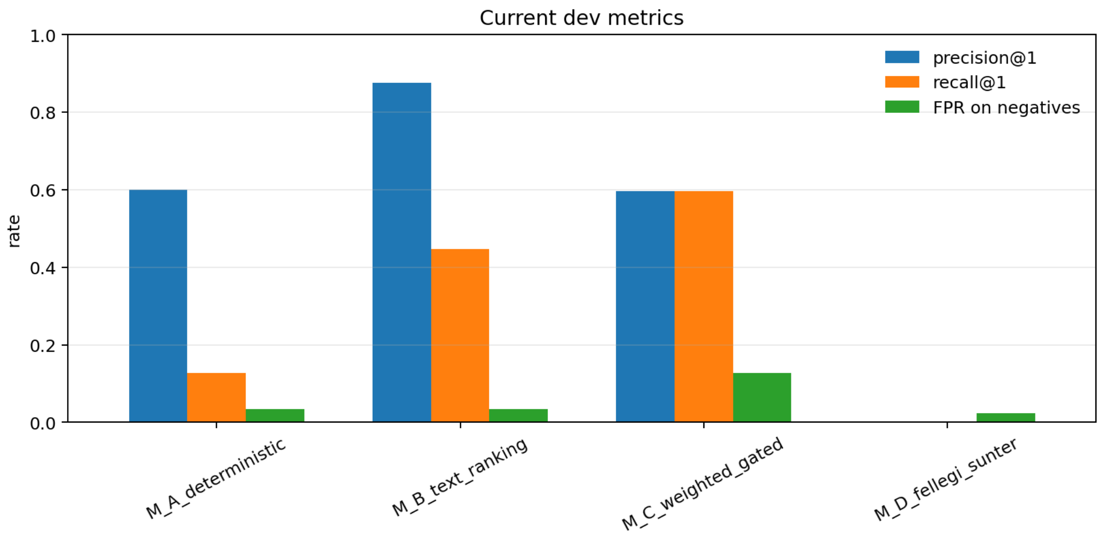

Figure 1: Dev split method comparison generated from the current benchmark evaluation JSON.

6 Validation Results

Validation is the method-selection evidence. The primary comparison is anchor-level exact-successor performance: each anchor can produce one accepted successor or abstain.

$$\text{precision} = \frac{TP}{TP + FP}, \quad \text{recall} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}.$$

For this project, false-positive control is prioritised because false links fabricate survival events.

Method	Threshold	Accepted	Precision	Recall	FPR	Coverage
M_A_deterministic	70.000	6	0.167	0.045	0.105	0.100
M_B_text_ranking	70.000	5	0.800	0.182	0.000	0.083
M_C_weighted_gated	70.000	15	0.600	0.409	0.105	0.250
M_D_fellegi_sunter	65.000	1	0.000	0.000	0.026	0.017

Table 2: Unweighted validation metrics on all labelled current benchmark frames.

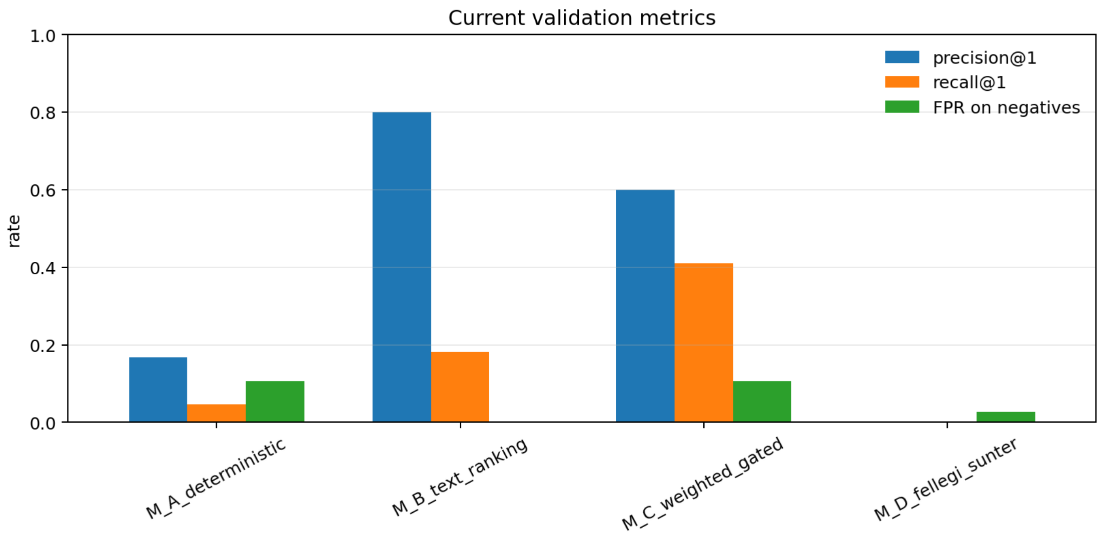

Figure 2: Validation split method comparison generated from the current benchmark evaluation JSON.

7 Quality Evidence And Interpretation

The validation result supports a conservative linkage claim. `M_B` has only 5 accepted validation links, so its precision estimate is necessarily sample-sensitive; one changed decision would move the number materially. However, the direction of the evidence is coherent: `M_B` gives the best validation precision and the lowest false-positive rate among useful methods, while `M_C` shows the expected precision-recall trade-off.

The pair-level ROC and precision-recall curves should be read as score-ranking diagnostics, not as the final operating decision. The final pipeline decision is anchor-level: choose one successor or abstain. The ROC curve is stepped rather than smooth because the validation benchmark contains a finite number of labelled pairs and many tied or near-tied scores.

8 Modeling Tables

The modeling-ready tables include strict, primary, broad, and non-match target columns, plus observable features. They now include the Fellegi–Sunter columns `fs_match_weight` and `fs_match_probability`, so modeling and evaluation use the same feature state.

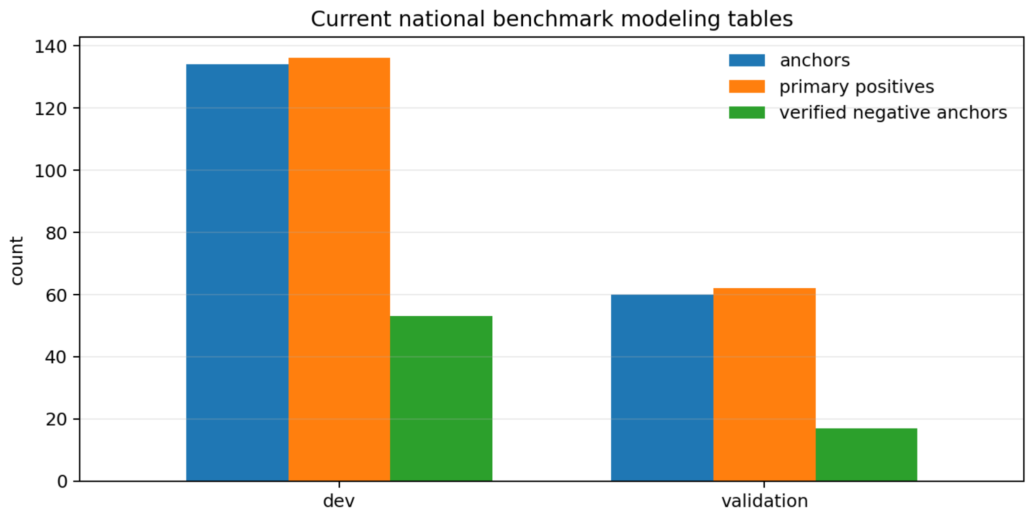

Figure 3: Current benchmark modeling table sizes and label support.

9 Survival Results

The survival dataset uses the main accepted links as events and all other episodes as right-censored observations at the study cutoff. The main variant contains 544 events and 3,256 censored observations. Segment event rates are 0.132 for CPV-32, 0.204 for CPV-35, 0.117 for CPV-48, and 0.143 for CPV-72. CPV-35 has the highest observed successor rate in the main arm.

Sensitivity checks show that absolute event rates depend strongly on the linkage rule: 296 events under `M_B @ 0.80`, 544 under `M_B @ 0.70`, 853 under `M_B @ 0.60`, and 1,332 under `M_C @ 0.70`. Therefore the survival estimates should be interpreted as linkage-conditioned descriptive evidence, not as exact renewal probabilities.

10 Defensible Decision

The current defensible decision is to keep `M_B_text_ranking @ 0.70` as the primary observable-successor method and use the stricter, looser, weighted-gated, and expiry-aware variants as sensitivity analyses. This is a precision-first design. Low recall is acceptable only because the report frames the event rate as a lower-bound proxy for visible reprocurement activity:

$$\text{observed successor rate} \leq \text{true renewal/reprocurement rate.}$$

If future manual review shows weaker precision, the claim should be narrowed further rather than forcing a more complex model.

11 Limitations And Robustness

- BOAMP does not consistently encode legal renewal status; accepted links are observable successor procurements, not legal renewal proof.
- The current benchmark labels are protocol-generated development evidence with double-pass validation, not independent human ground truth.
- Validation is small: 22 positive validation anchors and 5 accepted `M_B` validation links. The selected threshold should not be over-tuned.

- Buyer standardisation is improved but remains an important risk area. The legal-form audit is retained to catch name-only and cross-legal-form cases.
- Missing duration and expiry fields are not imputed. This avoids creating false timing certainty.

12 Recommended Reporting Position

The technical report should defend the project as a reproducible measurement pipeline, not as a black-box prediction system. The strongest statement is:

> Because legal renewal status is not directly observed in BOAMP, the study estimates observable successor procurements. Candidate generation restricts comparison to the same buyer and a plausible future interval; the selected method then accepts only the strongest textually similar candidate above a fixed threshold. This prioritises precision over recall, which is appropriate because false positive links would create artificial survival events.

13 Current Source Files

- `data/processed/boamp_v2/linkage_evaluation_summary_v3_dev_primary.json`
- `data/processed/boamp_v2/linkage_evaluation_summary_v3_validation_primary.json`
- `data/processed/boamp_v2/benchmark_v3/modeling/benchmark_v3_modeling_summary.json`
- `data/processed/boamp_v2/benchmark_v3/benchmark_v3_manifest.json`
- `data/processed/boamp_v2/survival_dataset_summary.json`
- `data/processed/boamp_v2/linkage_candidates_summary.json`